Example study report: watering schedules and seedling growth

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Summary

Three watering schedules were compared across 120 seedlings grown under identical light. Seedlings watered little and often grew taller than those watered once a week[1], and the difference held at both temperatures tested, 18 °C and 24 °C[2].

Mean height rose by **≥ 25%** under the daily schedule. Soil moisture in the daily trays stayed within ± 3% of target, and differences were significant at α = 0.05 ($p < 0.01$). Every schedule delivered 250 mL of water per tray per day on average.

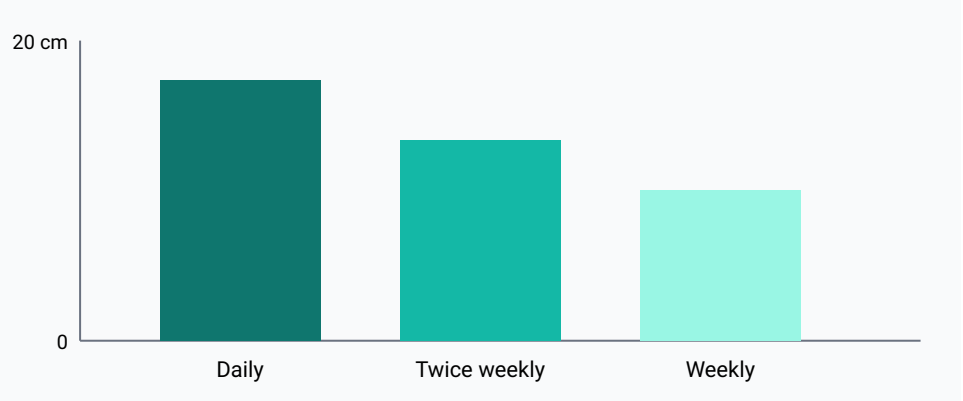
Key findings

- **Growth.** Daily watering produced the tallest seedlings by week 6[3].
- **Leaves.** Leaf count differed by at most one leaf between schedules.
- **Losses.** Two seedlings failed under the weekly schedule and none under the others.

Method

1. Seedlings were assigned at random to one of three schedules and one of two temperatures.
2. Height and leaf count were recorded every seven days.
3. Soil moisture was logged hourly by sensors in each tray.
4. Results were compared with a two-way analysis of variance, followed by pairwise tests between schedules.

Mean height at week 6



Schedules

Schedule	Volume per watering	Interval	Seedlings
Daily	250 mL	Every day	40
Twice weekly	875 mL	Every 3–4 days	40
Weekly	1,750 mL	Every 7 days	40

Schedule	Temperature	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Leaves at week 6	Source
Daily	18 °C	2.0 cm	4.4 cm	7.3 cm	10.4 cm	13.6 cm	16.9 cm	9	https://example.org/references/1
Daily	24 °C	2.2 cm	4.8 cm	7.8 cm	11.0 cm	14.3 cm	17.7 cm	9	https://example.org/references/1
Twice weekly	18 °C	1.9 cm	3.9 cm	6.1 cm	8.3 cm	10.6 cm	12.9 cm	8	https://example.org/references/2
Twice weekly	24 °C	2.0 cm	4.2 cm	6.5 cm	8.9 cm	11.3 cm	13.7 cm	8	https://example.org/references/2
Weekly	18 °C	1.8 cm	3.4 cm	5.0 cm	6.6 cm	8.1 cm	9.6 cm	8	https://example.org/references/3
Weekly	24 °C	1.9 cm	3.6 cm	5.3 cm	7.0 cm	8.7 cm	10.4 cm	8	https://example.org/references/3

Limitations

All seedlings came from a single seed batch and were grown in one facility, so the results may not hold for other varieties or conditions. Sensor readings drifted by up to 1% over the six weeks and were recalibrated weekly.

The study ran for six weeks only. Longer trials would show whether the early advantage of daily watering persists once plants are established.

References

[1] Example reference one: growth under frequent watering
Example Journal · 2024

[2] Example reference two: temperature and soil moisture
Example Journal · 2025

[3] Example reference three: measuring seedling height
Example Proceedings · 2026